Elements of AI

https://github.com/jendralhxr/introdatascience/blob/main/mathexpression.pdf

AI systems use mathematical/statistical expression (probability, variance, etc) to express uncertainty about the world instead of pretending to be sure.

Zulhaj Aliansyah

SEARCH TREES

Search trees: a simple yes/no decision

Question: should I take an umbrella? Each yes/no answer picks a branch; each path ends in a decision.

Is it raining?
YES

→

Decision
Take umbrella

Is it raining?
NO

→

Is it cloudy?
YES → Take umbrella
NO → Leave it at home

The tree lists the options; a search algorithm walks it from the root to a leaf.

SEARCH TREES

Route finding in Surabaya

Task: go from Gununganyar to Tunjungan Plaza. Every junction is a node, every road a branch. Times below are illustrative.

40 min

Route A

Gununganyar → Rungkut → Jemursari → Tunjungan

35 min

Route B

Gununganyar → MERR → Dharmahusada → Tunjungan

50 min

Route C

Gununganyar → Wonokromo → Tunjungan

The search keeps the cost so far on each branch and drops branches already worse than the best one found. Route C is pruned; B wins.

PROBABILITY

Probability: how likely is an event?

$0 \leq P \leq 1$

Definition

P(event) = favorable outcomes ÷ all possible outcomes. Always between 0 and 1.

$P(6) = 1/6$

Rolling a die

Six equally likely faces, one of them is a 6.

$P(\text{rain}) = 0.3$

From data

If it rained on 30 of 100 days, we estimate the chance of rain tomorrow.

AI systems use probability to express uncertainty about the world instead of pretending to be sure.

PROBABILITY

Compound probability, towards Bayes

Independent events

P(A and B) = P(A) × P(B)
Two heads: 0.5 × 0.5 = 0.25

Dependent events

P(A and B) = P(A) × P(B | A)
B’s chance changes once A has happened.

Case:

Population : 10,000 people.

Virulence : 1% of people *potentially* have the disease.

Test: catches 90% of real disease, but **also 5% on healthy person**.

How many actually potentially have health problem?

PROBABILITY

Dividing the population

- Healthy: 99% x 10000 = 9900 people
- Sick: 1% x 10000= **100 people**

- True Positive (sick, tested as sick)= 90% * 100 = 90 people
- False Positive (actually sick, tested as healthy)= 10% * 100= 10 people
- False Negative (appears healthy, tested as infected)= 5% * 9900= 495 people
- True Negative (healthy, tested as healthy)= 9900-891= 9009 people

People who should be cautious: 90+495= **545 people**

LINEAR REGRESSION

Fitting a line and measuring variance

The model

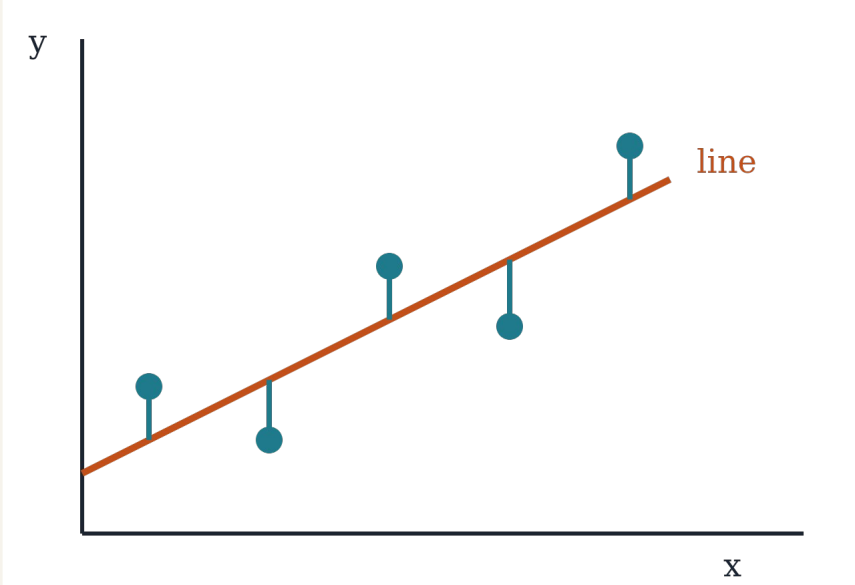
ŷ = w·x + b: predict y from x with one slope and one intercept.

Residual

y − ŷ: how far a point sits from the line.

Variance

Average squared distance from the mean. The best line minimises the squared residuals.



LINEAR REGRESSION

Real-life example: study time and exam score

Data (illustrative)

Hours studied: 1, 2, 3, 4, 5
Exam score: 55, 62, 70, 76, 85

$R^2 \approx 0.997$

Fitted line

score ≈ 47.4 + 7.4 × hours

Meaning

Each extra hour of study adds about 7 points on average.

≈ 91.8

Prediction

6 hours of study gives a predicted score of about 92.

R^2 tells how much of the variance in scores the line explains.

REGRESSION/INTERPOLATION IN UNSTRUCTURED DATA

Image upscaling with interpolation

Upscaling must invent new pixels between known ones. Cubic interpolation fits a smooth cubic curve through the neighbours: $p(t) = at^3 + bt^2 + ct + d$.

1

Nearest

Copies 1 pixel. Fast, blocky.

2

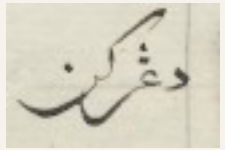
Linear

Mixes 2 pixels in 1D (4 in 2D). Smoother, a bit blurry.

4

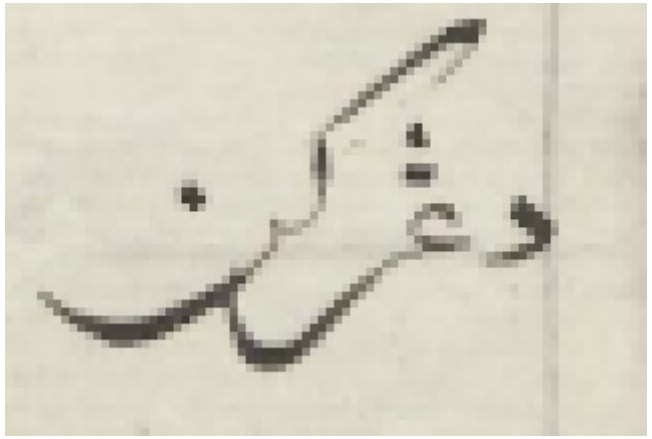
Cubic

Uses 4 pixels in 1D (16 in 2D, “bicubic”). Smooth with sharper edges.

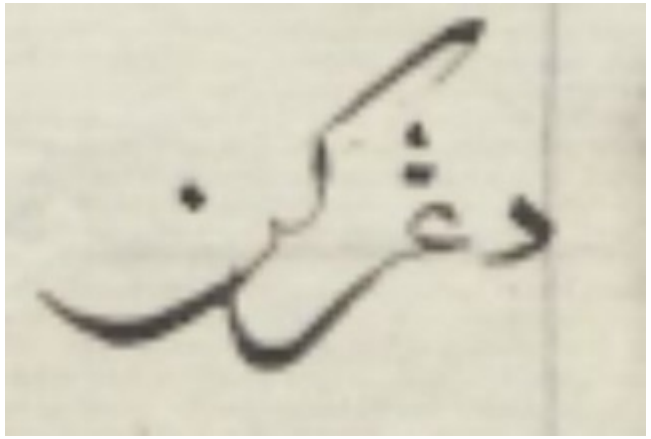


Source image (80x54 px)

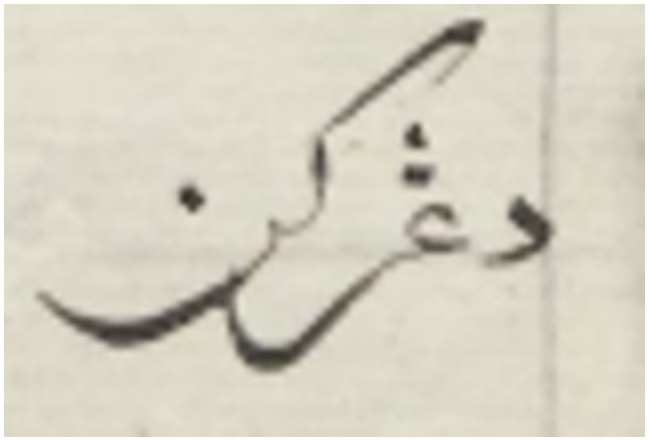
Nearest



Linear



Cubic



CLUSTERING

K-means: the rationale

Goal: put similar points in the same group, with no labels given. “Similar” means close, so we minimise the distance (variance) from each point to its group center/moment centroid.

1. Pick k

Choose how many groups, and place k starting centres.

2. Assign

Each point joins the nearest centre.

3. Update

Move each centre to the mean of its points.

4. Repeat

Stop when the assignments no longer change.

$$\arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 = \arg\min_{\mathbf{S}} \sum_{i=1}^{k} |S_i| \operatorname{Var} S_i$$

https://github.com/jendralhxr/introdatascience/blob/main/clustering.py

CLUSTERING · APPLICATION

Grouping shirt sizes from height and weight

Problem: we have people’s height and weight but no size labels.

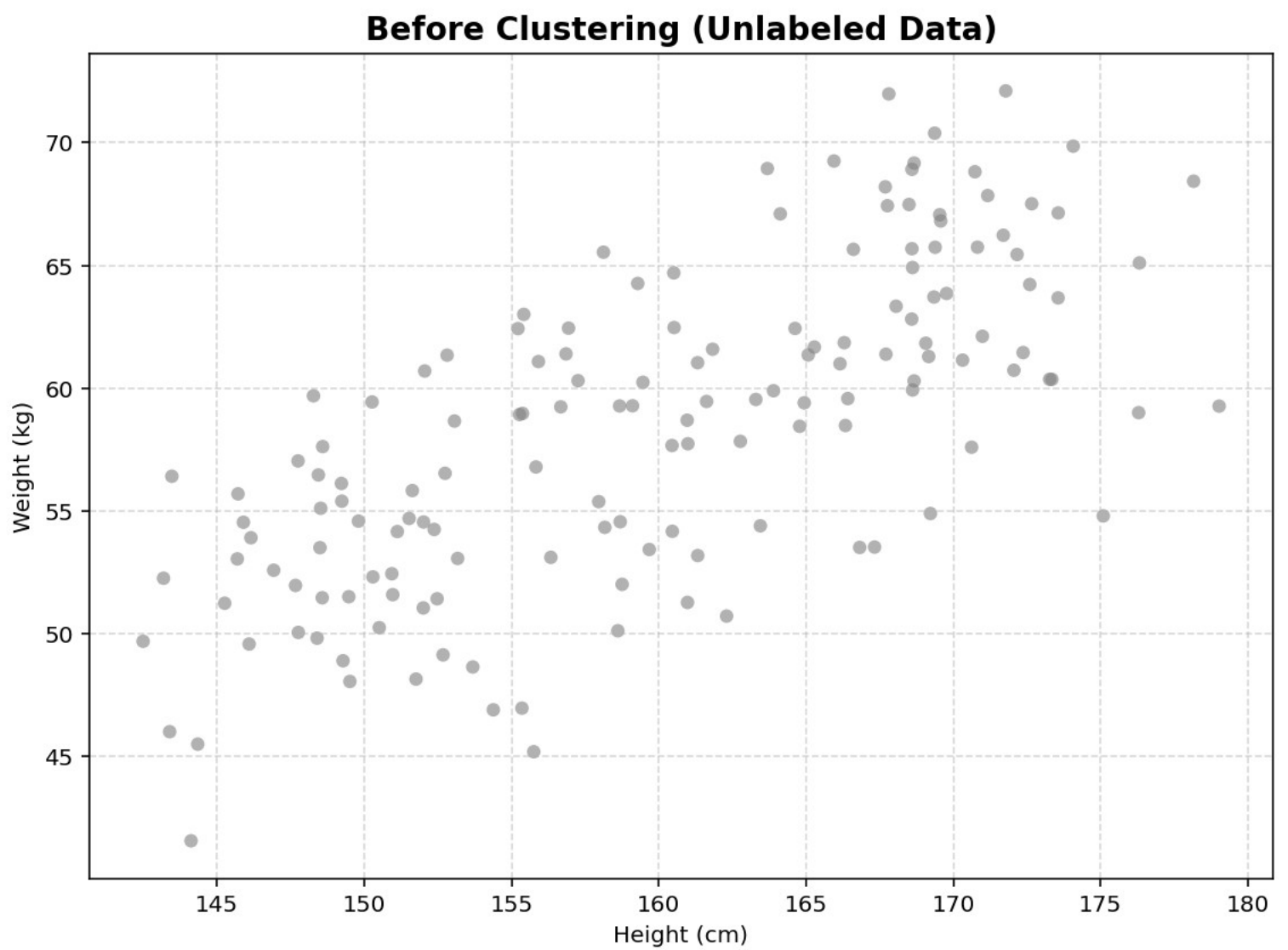
Cluster 1 → S

Cluster 2 → M

Cluster 3 → L

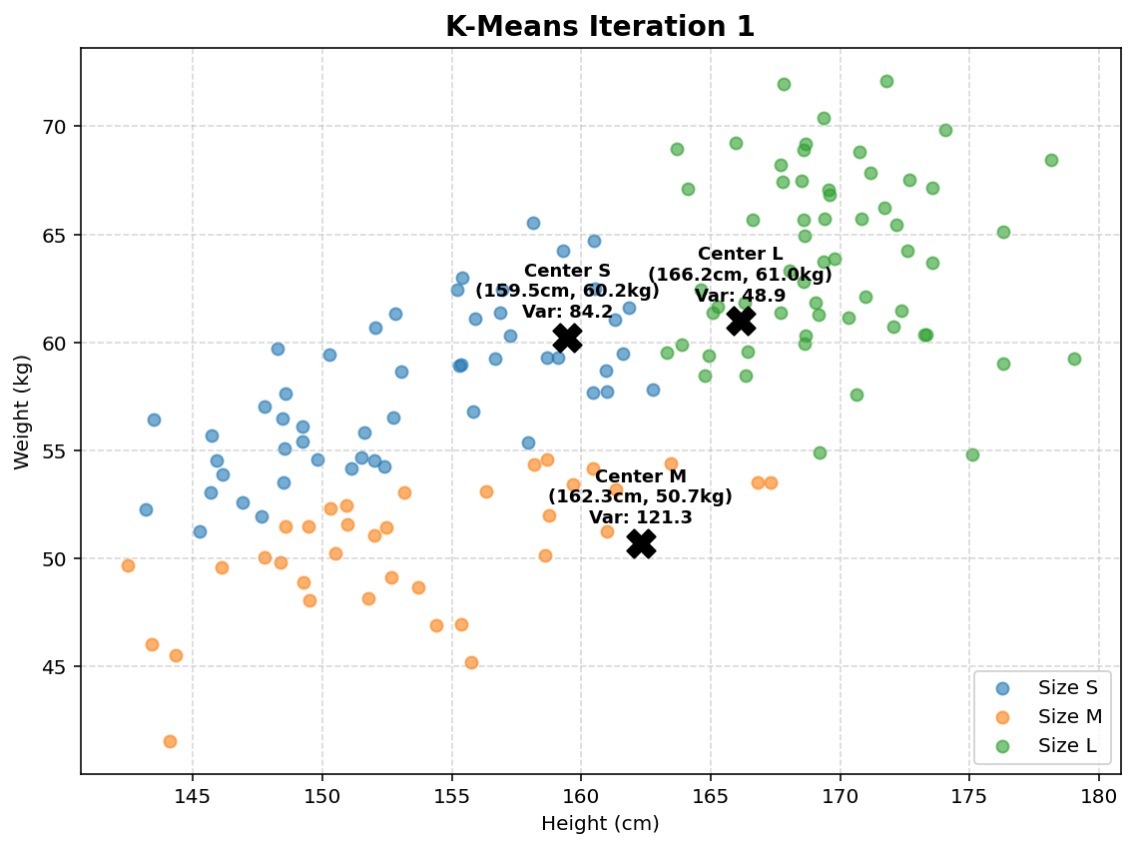
Each new person gets the size of the nearest centre. The choice of k (3, 4 or 5 sizes) is a business decision.

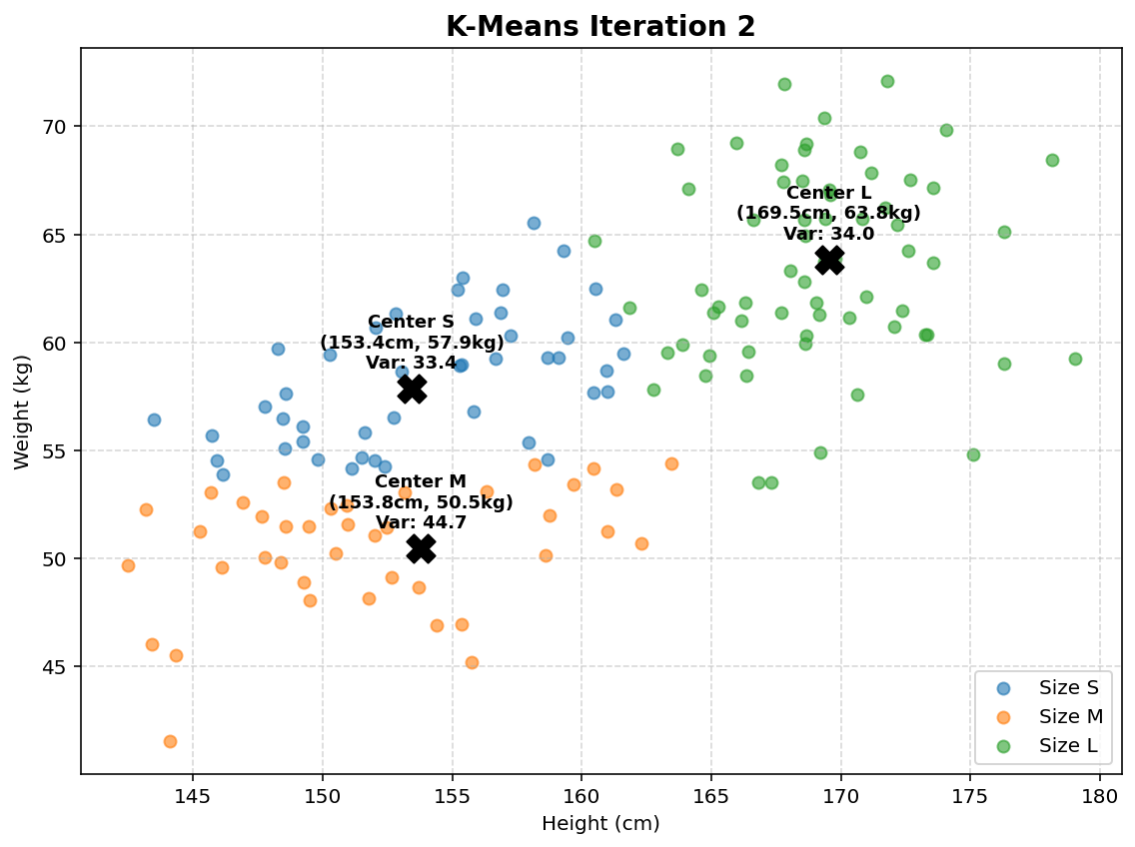
CLUSTERING



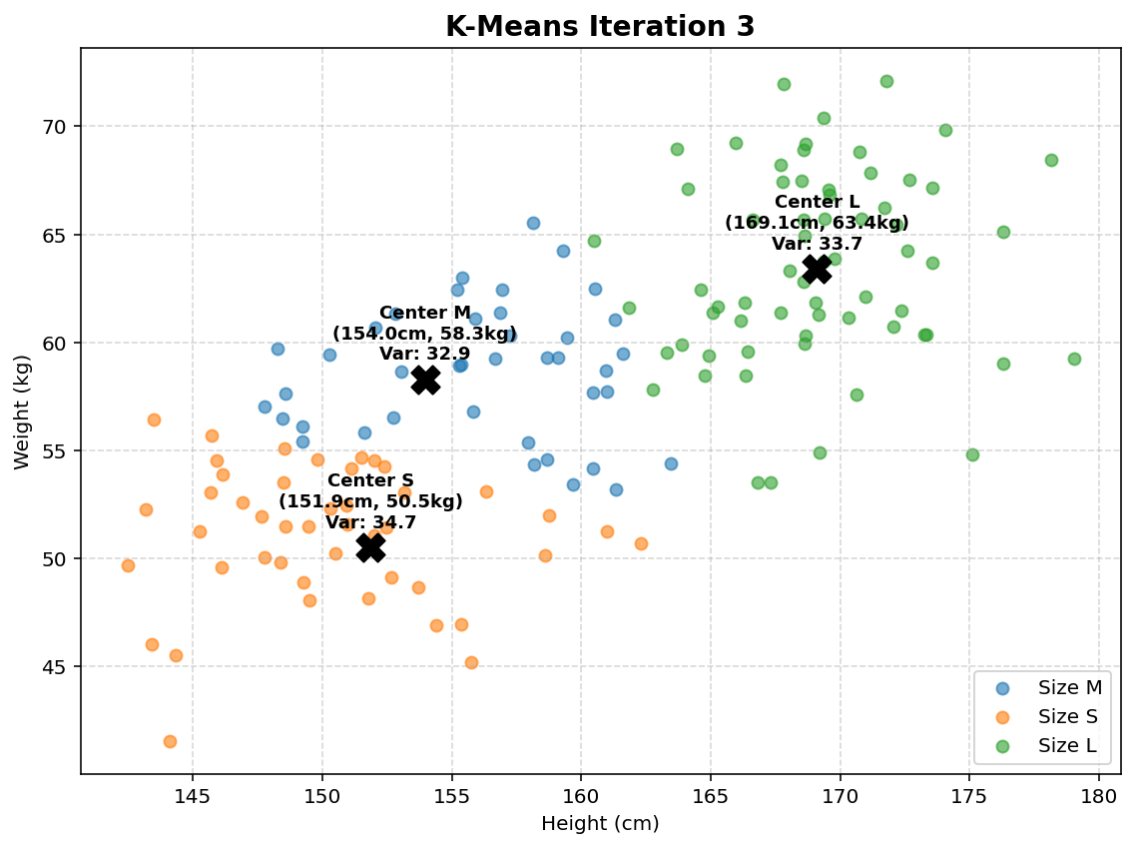
https://github.com/jendralhxr/introdatascience/blob/main/clustering.py

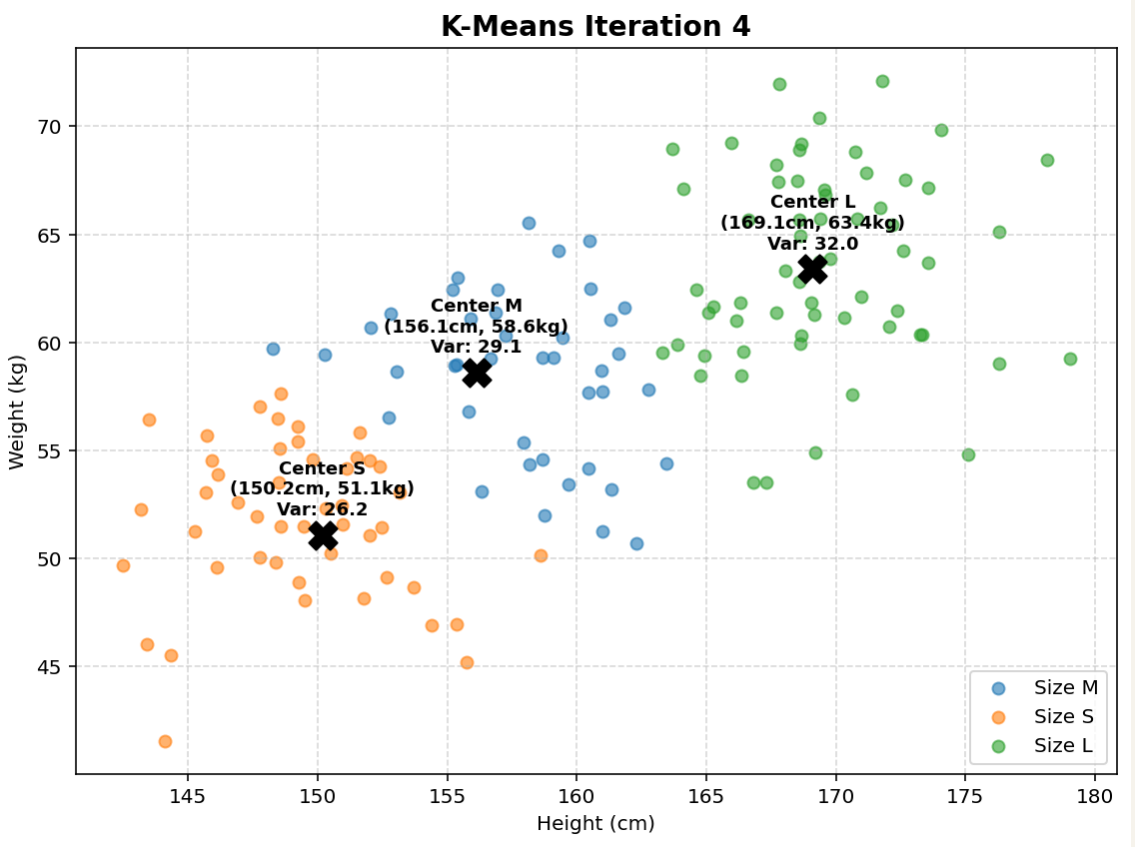
CLUSTERING


K-Means Iteration 1
Center S
(159.5cm, 60.2kg)
Var: 84.2
Center L
(166.2cm, 61.0kg)
Var: 48.9
Center M
(162.3cm, 50.7kg)
Var: 121.3
Size S
Size M
Size L
Weight (kg)
Height (cm)

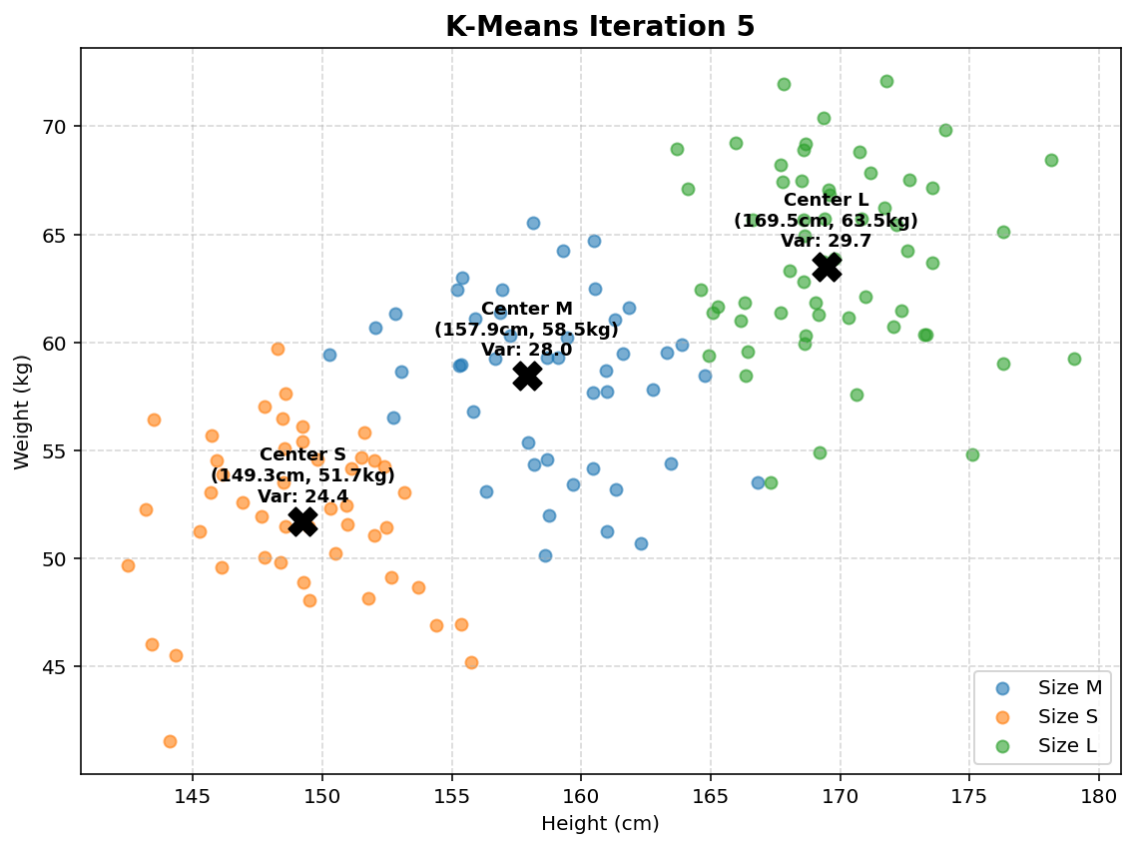

K-Means Iteration 2
Center L
(169.5cm, 63.8kg)
Var: 34.0
Center S
(153.4cm, 57.9kg)
Var: 33.4
Center M
(153.8cm, 50.5kg)
Var: 44.7
Size S
Size M
Size L
Weight (kg)
Height (cm)

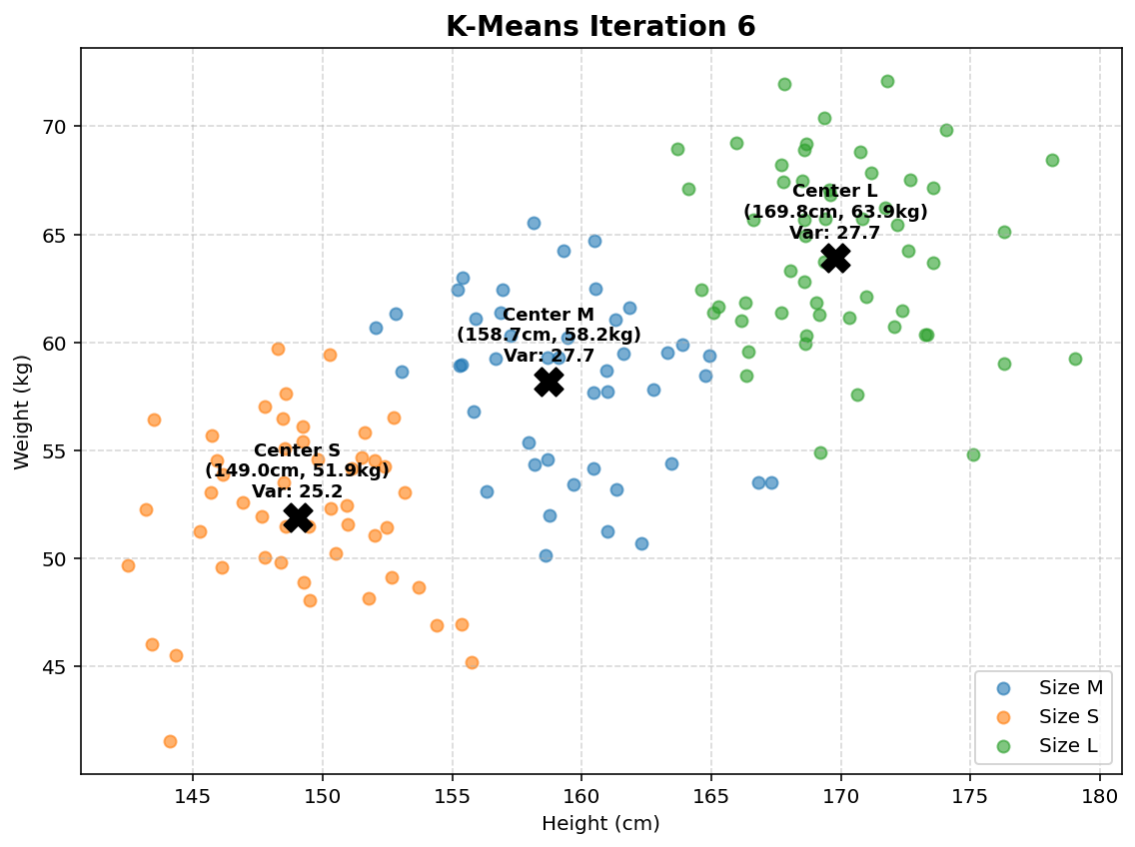
CLUSTERING


K-Means Iteration 3
Center L
(169.1cm, 63.4kg)
Var: 33.7
Center M
(154.0cm, 58.3kg)
Var: 32.9
Center S
(151.9cm, 50.5kg)
Var: 34.7
Size M
Size S
Size L
Weight (kg)
Height (cm)
70
65
60
55
50
45
145
150
155
160
165
170
175
180

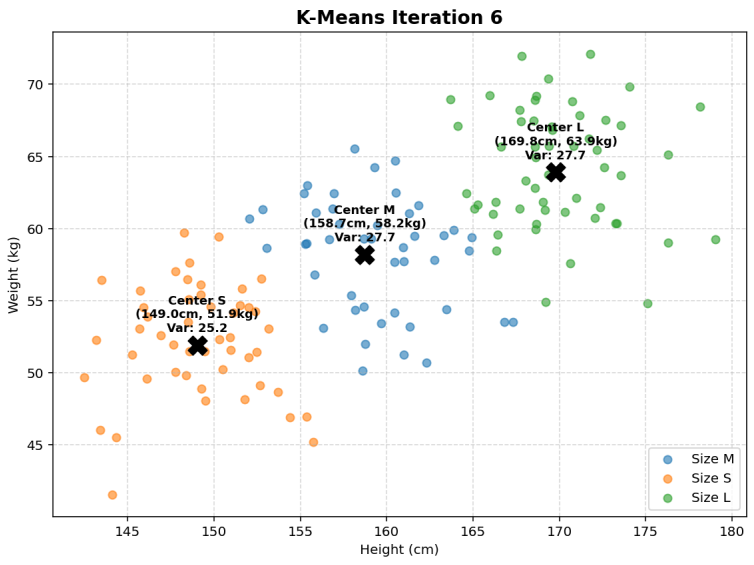

K-Means Iteration 4
Center L
(169.1cm, 63.4kg)
Var: 32.0
Center M
(156.1cm, 58.6kg)
Var: 29.1
Center S
(150.2cm, 51.1kg)
Var: 26.2
Size M
Size S
Size L
Weight (kg)
Height (cm)
70
65
60
55
50
45
145
150
155
160
165
170
175
180

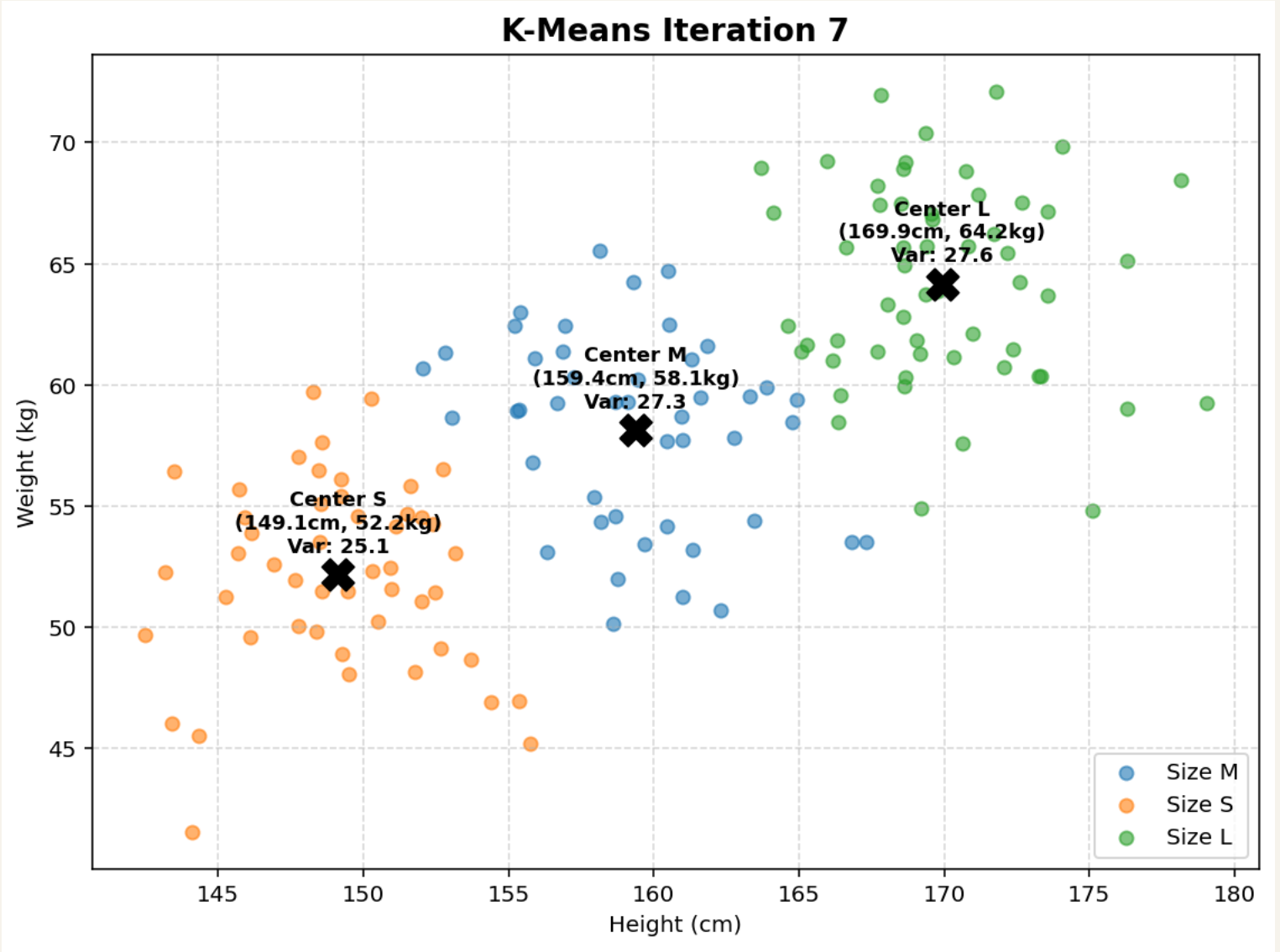
CLUSTERING


K-Means Iteration 5
Center L
(169.5cm, 63.5kg)
Var: 29.7
Center M
(157.9cm, 58.5kg)
Var: 28.0
Center S
(149.3cm, 51.7kg)
Var: 24.4
Size M
Size S
Size L
Weight (kg)
Height (cm)
70
65
60
55
50
45
145
150
155
160
165
170
175
180

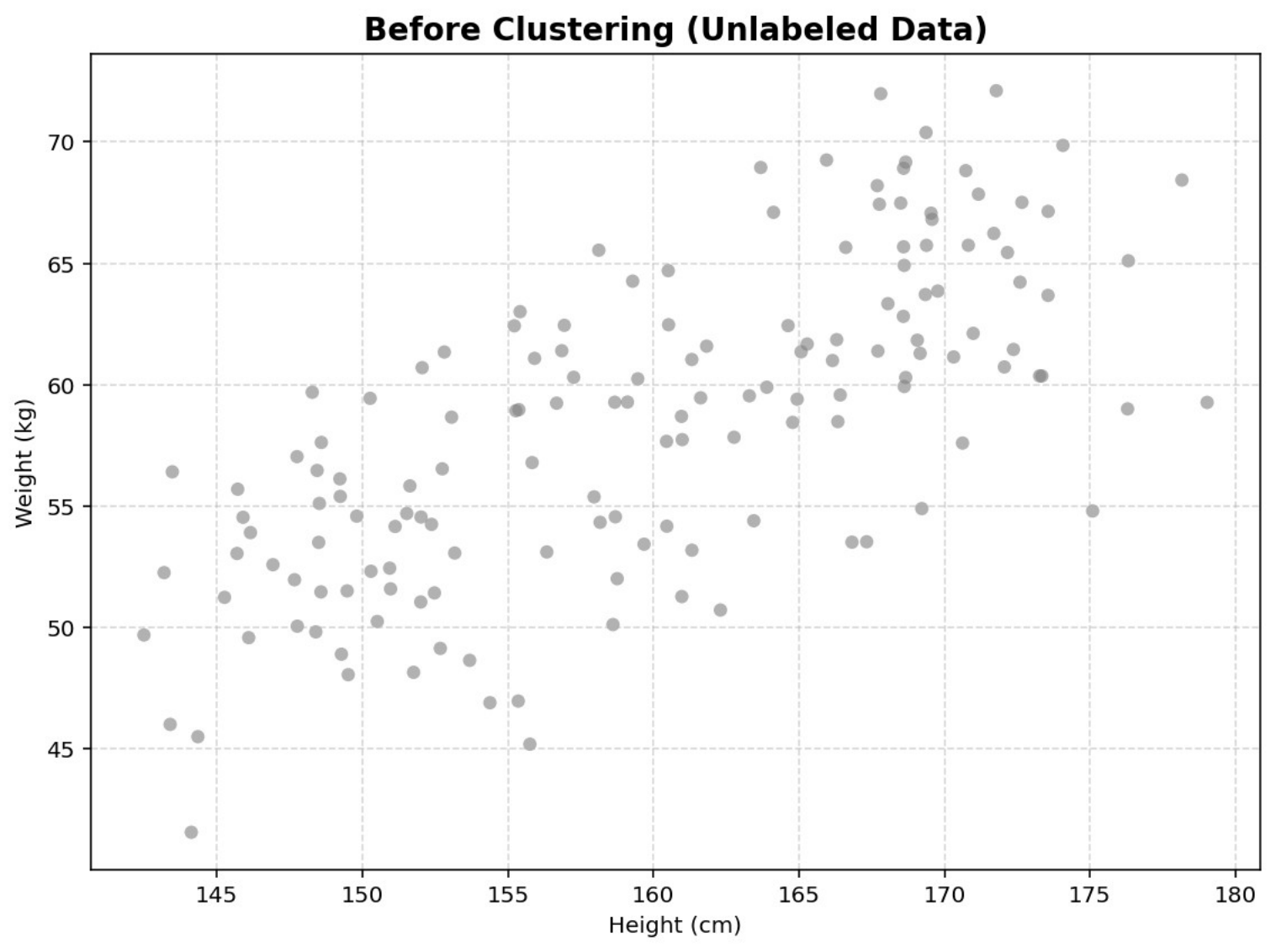

K-Means Iteration 6
Center L
(169.8cm, 63.9kg)
Var: 27.7
Center M
(158.7cm, 58.2kg)
Var: 27.7
Center S
(149.0cm, 51.9kg)
Var: 25.2
Size M
Size S
Size L
Weight (kg)
Height (cm)
70
65
60
55
50
45
145
150
155
160
165
170
175
180

CLUSTERING


K-Means Iteration 6
Center L
(169.8cm, 63.9kg)
Var: 27.7
Center M
(158.7cm, 58.2kg)
Var: 27.7
Center S
(149.0cm, 51.9kg)
Var: 25.2
Size M
Size S
Size L
Weight (kg)
Height (cm)
70
65
60
55
50
45
145
150
155
160
165
170
175
180


K-Means Iteration 7
Center L
(169.9cm, 64.2kg)
Var: 27.6
Center M
(159.4cm, 58.1kg)
Var: 27.3
Center S
(149.1cm, 52.2kg)
Var: 25.1
Size M
Size S
Size L
Weight (kg)
Height (cm)
70
65
60
55
50
45
145
150
155
160
165
170
175
180

CLUSTERING


Before Clustering (Unlabeled Data)
Weight (kg)
70
65
60
55
50
45
145
150
155
160
165
170
175
180
Height (cm)

CLUSTERING · RESULT

Grouping shirt sizes from height and weight

Problem: we have people’s height and weight but no size labels.

Cluster 1 → S
About 149 cm, 52 kg

Cluster 2 → M
About 159 cm, 58 kg

Cluster 3 → L
About 169 cm, 64 kg

Each new person gets the size of the nearest centre. The choice of k (3, 4 or 5 sizes) is a business decision.